

# MOHAMMED AMAN BHUIYAN

🏠 Dhaka, Bangladesh    📞 (+880) 1915-003602    ✉️ mohammed.aman.nsu@gmail.com  
🌐 mohammedaman.vercel.app    📺 mohammedaman    🎧 Aman554-EQ    📺 Aman554

## PROFILE STATEMENT

AI/ML researcher and aspiring Machine Learning Engineer with expertise in Natural Language Processing, Large Language Model optimization (Pruning, Quantization), Automatic Speech Recognition (ASR), and Computer Vision for action recognition and temporal localization. Experienced in building efficient, scalable models with a focus on edge deployment and inference optimization. Committed to advancing research in LLMs/SLMs and video understanding by developing practical, high-impact solutions grounded in strong theoretical foundations.

## EXPERIENCE

### North South University (NSU)

Feb '24 – Present

*Undergraduate Teaching Assistant (Department of ECE)*

*Dhaka, Bangladesh*

- Co-taught **CSE373: Design & Analysis of Algorithms** (Undergraduate), mentoring 100+ students in algorithmic problem-solving, complexity analysis, and programming techniques.
- Currently instructing **CSE445: Machine Learning** (Graduate level), guiding hands-on model implementation, and supporting research-oriented coursework.
- Designed assignments, evaluated coursework, and held consultation hours to improve students conceptual clarity and academic performance.

### Research Project – Classroom Monitoring Technology

Jan '25 – Sep '25

*(Worked Under the supervision of Dr. Shafin Rahman)*

*North South University*

- Developed an **end-to-end AI pipeline** aligned with thesis research for automated classroom analytics using real-world video data.
- Trained and optimized deep learning models on a proprietary classroom dataset provided by the supervisor.
- Built a full-stack application that processes classroom video input and generates comprehensive engagement reports.
- Integrated data preprocessing, model inference, evaluation metrics, and reporting into a production-ready system.

## EDUCATION

### North South University

Sep '21 – Dec '25

*Bachelor of Science in Computer Science and Engineering, CGPA: 3.68(Magna Cum Laude)*

*Dhaka, Bangladesh*

## TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C++
- **AI & ML Tools:** PyTorch, HuggingFace, Scikit-learn, LangChain, LlamaIndex, Ollama
- **Data Manipulation & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Web Development:** FastAPI, Flask, MERN Stack, Jest, JSDoc
- **Databases:** MySQL, MongoDB, PostgreSQL, Redis, VectorDB (FAISS, ChromaDB, Qdrant)
- **Other Expertise:** Git, Docker, REST API Development, Data Analysis, Web Scraping, RAG Pipelines, LLM/SLM Fine-tuning, Model Compression, Prompt Engineering

## PROJECTS

### SHOBDOTORI: Regional-Dialect-to-Standard Bangla Speech Recognition

🎧 ShobdoTori\_ASR

*Python, PyTorch, Whisper, Wav2Vec2, KenLM, ASR*

- Developed an Automatic Speech Recognition (ASR) system to transcribe regional Bangladeshi dialects into standard Bangla.
- Curated a **3,800+** audio-sample dataset covering **20 regional** dialects across Bangladesh.
- Implemented Transformer-based ASR (**Whisper, Wav2Vec2**) with phoneme alignment and audio data augmentation.
- Applied **n-gram KenLM** post-processing to improve transcription fluency and linguistic correctness.
- Evaluated performance using Normalized Levenshtein Similarity (**NLS**), achieving competitive leaderboard results.

### BanglaWhisperDiar: Long-Form ASR & Speaker Diarization for Bangla

🎧 BitwiseMind\_ASR\_Diar

*Python, PyTorch, Whisper, PyAnnote, PyTorch Lightning, librosa*

- Built a dual-pipeline deep learning system for long-form Bengali conversational audio transcription and speaker diarization.
- Curated a **15,000+** aligned audio-segment dataset using sequential fuzzy text matching with active transcription, and applied stochastic audio augmentations to improve model robustness.
- Fine-tuned the **Whisper-Medium** ASR model using 8-bit AdamW optimization and mixed-precision training for efficient large-scale training.

- Optimized long-audio inference with **FasterWhisper CT2Translate**, processing a 24+ hour test set in **25 minutes** (Only **1 minute** per audio file).
- Fine-tuned the **PyAnnote Segmentation 3.0** model using PyTorch Lightning and integrated it into a customized speaker diarization pipeline.
- Evaluated cross-domain generalization using Word Error Rate (WER) and Diarization Error Rate (DER) as benchmarking metrics, achieving a **28.4%** improvement in WER and a **40.3%** improvement in DER compared to the baseline.

### TimeClipAI: Real-Time Action Classification and Time Segmentation

 TimeClip\_AI

*PyTorch, Anchor Transformers, I3D, Python, Git*

- Developed **TimeClipAI**, a framework for **online temporal action localization** using Anchor Transformers
- Enabled **real-time action classification and time segmentation** in full-length videos
- Integrated support for multiple benchmark datasets including EGTEA, EPIC-Kitchen 100, THUMOS'14, and CricShot10
- Leveraged pre-trained **I3D features** for efficient training and testing
- Designed and trained a **post-processing Offset Scoring Network (OSN)** to refine predictions
- Achieved **state-of-the-art** performance with robust generalization across datasets

### Educational Chatbot using RAG

 Physics\_RAG\_Chatbot

*SLMs, Python, Ollama, LangChain, HuggingFace, FastAPI, Prompt Engineering, Git*

- Developed a lightweight Retrieval-Augmented Generation (RAG) chatbot using small language models (SLMs) for offline and efficient educational assistance
- Built pipeline with **LangChain**, **Ollama**, and **HuggingFace** for document retrieval, embedding, and response generation
- Evaluated performance on a **5,000+** self-curated physics Q&A dataset, achieving **75%** response accuracy using **RAGChecker** framework
- Designed a user-friendly web interface with **FastAPI** to improve accessibility for students in low-resource environments

### Road Accident Analysis System

 RoadRisk\_ClassicML

*Python, Scikit-learn, NumPy, Pandas, Git*

- Built a machine learning system to predict patient status and injury severity using a Road Traffic Accident dataset
- Performed **data preprocessing, feature engineering**, and model evaluation using Scikit-learn and Pandas
- Implemented and compared multiple models including Logistic Regression, Decision Tree, SVM, Random Forest, & XGBoost
- Achieved **99.1%** accuracy in patient status prediction and **89%** accuracy in injury type classification

### Agile-Driven Movie Reservation Website

 CholoDekhi.327

*Node.js, React, MongoDB, Postman, Jest, Trello, GitHub Wiki, JSDoc, Git*

- **Led** a 5-member team using **Agile Scrum** to develop a movie reservation platform
- Contributed to both frontend and backend following **MVC** architecture
- Organized and maintained project documentation using **GitHub Wiki** and **JSDoc**
- Coordinated **sprint tasks** and ensured milestone delivery within planned timelines
- Improved reliability through unit testing with **Jest**

## Publications

### Z-Pruner: Post-Training Pruning of Large Language Models for Efficiency without Retraining

Samiul Basir Bhuiyan, Md. Sazzad Hossain Adib, **Mohammed Aman Bhuiyan**, Muhammad Rafsan Kabir, Moshir Farazi, Shafin Rahman, and Nabeel Mohammed

Accepted at **IEEE AICCSA 2025**  [arxiv.org/abs/2508.15828](https://arxiv.org/abs/2508.15828)

## Extra-Curricular Activities & Competitions

### National AI Competition Finalist, AI Fication 2025 (Televerse 1.0)

*Chittagong University of Engineering & Technology (CUET)*

- Focused on regional-to-standard Bangla speech transcription
- Ranked **4th** in the private online round and **6th overall** in the final stage

### Finalist (out of 192+ teams), BUET AI HACKATHON DL Sprint 4.0

*Bangladesh University of Engineering & Technology (BUET)*

- Built solutions for Bangla long-form audio ASR transcription and speaker diarization

### Top 15 Finalist (out of 100+ teams), NSU Capstone Project Challenge – Innovation Challenge 2025

*North South University (NSU)*

- Recognized for developing impact-driven and innovative project solutions

## References

### Dr. Shafin Rahman

Associate Professor, North South University

✉ [shafin.rahman@northsouth.edu](mailto:shafin.rahman@northsouth.edu)

### Dr. Nabeel Mohammed

Associate Professor, North South University

✉ [nabeel.mohammed@northsouth.edu](mailto:nabeel.mohammed@northsouth.edu)